COMPREHENSIVE PRODUCT SPECIFICATION

ROYALIS LUXURY COMMERCE platform

Next-Generation E-Commerce Engine Architecture & High-Performance Front-End Specification

BRAND NAME:	ROYALIS (Evolved from "Royals")
TAGLINE:	The Sovereign Standard of Commerce
CORE NICHE:	Curated Hyper-Luxury, Premium Tech, Direct Artisan & VIP Concierge Services
TECH STACK:	Vanilla HTML5, Modern CSS3 (Grid/Flexbox/Variables), Native JavaScript (ES6+)
AUTHOR:	Lead Systems Architect & UI/UX Director
TARGET COMPLETION:	Single-Day Execution Protocol (Today)

Executive Summary & Vision Statement

Royalis is designed to outclass legacy global marketplaces like Amazon by replacing infinite noise with uncompromised quality, ultra-premium UI/UX design, and AI-driven hyper-personalization. While Amazon excels in sheer volume and fast logistics, Royalis focuses on the high-margin, elite tier of global commerce—combining authenticated premium goods, immersive 3D/interactive product showcases, instant VIP concierge assistance, and frictionless zero-friction checkout.

This document details the complete end-to-end blueprint for building the front-end interface using pure HTML, CSS, and JavaScript without external library overhead, ensuring lightning-fast performance, zero render blocking, and sovereign visual presentation.

1. Brand Identity & Competitive Differentiators

To surpass Amazon in quality, customer loyalty, and prestige, Royalis abandons cluttered ads and cheap product listings in favor of bespoke luxury and flawless design.

1.1 Naming & Positioning Architecture

BRAND ATTRIBUTE	AMAZON MODEL	ROYALIS MODEL (NEXT-GEN SUPERIORITY)
Brand Name & Vibe	Mass-market marketplace (Amazon)	**ROYALIS** — Sovereign, prestigious, uncompromising luxury.
Curated Value Prop	Millions of unverified 3rd-party items; heavy spam & fake reviews.	**Strict Authenticity & Quality Curation**: Only top-tier, verified, premium goods and direct artisan brands.
Customer Experience	Ad-saturated, transactional, cluttered list layouts.	**Cinematic & Immersive UI**: High-resolution visual cards, dark-mode elegance, micro-interactions, clean focus.
Logistics & VIP Care	Standard automated support bots and delayed email help.	**Royalis Sovereign Concierge**: 1-click live VIP concierge, tailored recommendations, real-time live map tracking.
Speed & Architecture	Heavy tracking scripts, multi-second loading bloat.	**Zero-Framework Vanilla JS Engine**: Under 100ms UI responses, smooth animations, native browser speed.

2. Pure Tech Stack & Design System

The technical constraint for today is strictly **HTML5, CSS3, and Native ES6+ JavaScript**. No npm packages, react frameworks, or bulky stylesheets will be used, guaranteeing pure performance.

2.1 Visual Tokens & CSS System Variables

All CSS properties utilize CSS Variables to ensure seamless instant theme switching (Obsidian Dark vs. Pearl Light) and modular color management.

```
:root {
  /* Royalis Gold & Sapphire Theme Palette */
  --bg-primary: #0f172a;        /* Deep Obsidian Navy */
  --bg-secondary: #1e293b;      /* Sleek Card Surface */
  --bg-tertiary: #182234;       /* Subtle Alt Background */
  --accent-gold: #f59e0b;       /* Sovereign Gold (CTAs, Headers) */
  --accent-gold-hover: #d97706; /* Darker Gold State */
  --accent-cyan: #38bdf8;       /* Futuristic Accent */
  --text-primary: #f8fafc;      /* Pure White Text */
  --text-muted: #94a3b8;        /* Soft Slate Text */
  --border-light: rgba(255, 255, 255, 0.1);
  --shadow-elevation: 0 10px 30px -10px rgba(0, 0, 0, 0.5);
  --font-heading: 'Plus Jakarta Sans', system-ui, sans-serif;
  --font-body: 'Inter', system-ui, sans-serif;
}
```

2.2 Typography Hierarchy

- **H1 (Hero Displays):** 32pt - 40pt / Bold 800 / Tracking -1px
- **H2 (Section Titles):** 18pt - 22pt / SemiBold 700 / Accent Gold Border
- **H3 (Card Titles):** 12pt - 14pt / Medium 600 / Clean White
- **Body Text:** 9.5pt - 10.5pt / Regular 400 / Line Height 1.6

3. Feature Breakdown & Priority Backlog

To complete the build today, features are divided into strict priority tiers (P0 = Critical MVP, P1 = Essential Enhancements, P2 = Amazon-Killer Innovations).

PRIORITY	FEATURE MODULE	TECHNICAL REQUIREMENT & SCOPE	STATUS
P0	**Navigation Header**	Sticky blurred Glassmorphic header with Logo, Instant Search Input, Category Drops, Wishlist counter, and Cart Badge.	Build Today
P0	**Interactive Product Grid**	Responsive 4-column layout rendering dynamic product cards with star ratings, quick badges, price, and instant modal view.	Build Today
P0	**Slide-Out Cart Drawer**	Pure JS slide-in drawer with live quantity increment/decrement, tax/shipping computation, and checkout trigger.	Build Today
P0	**Product Detail Modal**	Overlay modal showing high-res gallery, size/color variant switches, stock indicator, and specifications accordion.	Build Today
P1	**Dynamic Multi-Filter**	Client-side Javascript filtering by category, price slider, in-stock state, and rating without page refresh.	Build Today
P1	**Sovereign Checkout Flow**	Accordion checkout form with real-time card validation, order summary calculations, and confirmation modal.	Build Today
P2	**AI Concierge Assistant Widget**	Interactive floating chat modal for personal shopping advice and direct product queries.	Build Today

4. Single-Day Implementation Roadmap

Follow this exact hourly execution plan to complete the full prototype by the end of the day using your AI coding environment (Cursor / V0 / VS Code).

> **Phase 1: Project Setup & Core Shell (Hours 1–2)**
>
> Create `index.html`, `styles.css`, and `app.js`. Establish CSS Variables, layout container max-widths, and the responsive navigation bar with glassmorphic backdrop blur (`backdrop-filter: blur(12px)`).

Phase 2: Hero Section & Dynamic Catalog Data (Hours 3–4)

In `app.js`, create a mock product database array containing 8–12 ultra-luxury items (e.g., Quantum Watches, Hyper-EV Speedsters, Artisan Silk Wear) with titles, tags, prices, ratings, and image links.

Phase 3: JavaScript State Management & Cart Drawer (Hours 5–6)

Implement `cartState` array in JavaScript. Write functions for `addToCart(productId)`, `updateQuantity(productId, delta)`, `renderCart()`, and `calculateTotals()`. Bind event listeners to sliding drawer toggles.

Phase 4: Filtering Engine & Product Detail Modal (Hours 7–8)

Connect search input and filter checkboxes to a master `filterProducts()` function. Build the modal trigger for detailed product view, variant picking, and instant buy features.

Phase 5: Sovereign Checkout & UI Polish (Hours 9–10)

Build the checkout modal with dynamic itemized pricing. Test mobile responsiveness, hover effects, subtle gold shadows, and smooth scrolling.

5. Starter Code Architecture

5.1 JavaScript Product Data Engine (`app.js`)

```
// Product State & Catalog Engine
const products = [
  {
    id: 'roy-001',
    name: 'Royalis Chrono-Sovereign Tourbillon',
    category: 'Horology',
    price: 48500,
    rating: 4.95,
    badge: 'Curated Gold',
    image: 'https://images.unsplash.com/photo-1523275335684-37898b6baf30?w=600',
    description: 'Precision grade-5 titanium tourbillon timepiece featuring sapphire glass casing and
72-hour power reserve.'
  },
  {
    id: 'roy-002',
    name: 'Aero-Luxe Cyber Acoustic Headphones',
    category: 'Audio',
    price: 2400,
    rating: 4.88,
    badge: 'Limited Supply',
    image: 'https://images.unsplash.com/photo-1505740420928-5e560c06d30e?w=600',
    description: 'Bespoke planar magnetic headphones wrapped in fine Italian leather with active
acoustic cancellation.'
  }
];
```

```
let cart = [];

function addToCart(productId) {
  const product = products.find(p => p.id === productId);
  const existing = cart.find(item => item.id === productId);
  if (existing) {
    existing.qty += 1;
  } else {
    cart.push({ ...product, qty: 1 });
  }
  updateCartUI();
}

function updateCartUI() {
  const cartBadge = document.getElementById('cart-badge');
  const totalItems = cart.reduce((sum, item) => sum + item.qty, 0);
  if (cartBadge) cartBadge.textContent = totalItems;
}
```